Jack Hedin's Project

Dr. Stuart Jefferies

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Research Goals

- Determine the error bars for the plot of "peak detection SNR" versus "companion magnitude"
- Determine how bright an unresolved object needs to be before it's spectrum can be faithfully reproduced (i.e., spectral angle < 10 degrees) using broadband speckle data with: a) direct restoration, and b) joint linear regression. Perform a Monte Carlo to generate error bars,
- Determine the sensitivity of the above measurements to PSF fidelity
- Compare restorations made with weighted least-squares and 2-D/3-D maximum residual likelihood
- Look at two-unresolved sources problem addressed with joint linear regression

Sub-project #1

- Use a point source object